

AeroTwin

Real-Time Digital Twin for Predictive Aircraft Engine Health Monitoring

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→ AEROSPACE MRO

🔧 EDGE AI + ML

🔧 DIGITAL TWIN

⚠️ THE PROBLEM

Aviation MRO today runs on fixed inspection schedules — ground an aircraft on a calendar date, not on actual engine health. This creates two costly extremes: unnecessary groundings when parts are still healthy, and missed faults when degradation outpaces the schedule. A single unplanned AOG event costs airlines \$150,000+ per day. The global engine MRO market is \$33B annually — with 20–30% of costs attributable to inefficient scheduled maintenance.

🔧 OUR SOLUTION

AeroTwin is a real-time Digital Twin system that creates a continuously-updated virtual replica of an aircraft engine. Live sensor telemetry streams into a fatigue-calculation engine and an ML model that predicts Remaining Useful Life (RUL) per component — giving maintenance teams continuous, actionable insight instead of periodic snapshots.

🔄 HOW IT WORKS

- ◆ 1. Sensor simulation engine generates realistic temperature, vibration & RPM data
- ◆ 2. Fatigue algorithm scores component health every simulation tick
- ◆ 3. Random Forest ML model predicts RUL with confidence interval
- ◆ 4. WebSocket streams all data to React dashboard in real time
- ◆ 5. Alert engine fires tiered maintenance recommendations instantly

< 2 sec Anomaly detection & alert latency

~12–18 hrs ML model RUL prediction MAE

20–30% Estimated reduction in unnecessary inspections

10–15% Potential extension of component useful life

\$33B+ Global engine MRO market opportunity

3 components Monitored simultaneously (turbine, bearing, compressor)

Frontend: React.js · Three.js · Recharts

Backend: Python · Flask · Flask-SocketIO

ML: scikit-learn (Random Forest)

DB: SQLite

Comms: Socket.IO WebSocket

✓ Local inference — no cloud dependency

✓ Sub-100ms update latency end-to-end

✓ Full offline operability (air-gap ready)

✓ Compressed ML model (edge-hardware ready)

✓ Reduced data transmission (insights only)

Prakash Tiwari (Lead) Architecture, Backend, ML

Harnoor Kaur Frontend, UI/UX, Demo

Prince Sagwal Simulation Engine, DB

Piyush Kumar ML Pipeline, Documentation

📄 NASA-Calibrated
Simulation Engine

⚡ Real-Time Anomaly
Detection < 2 sec

📄 Open-Source &
Accessible Architecture

Team FELONS | AeroTwin | InnoVent-27 | AI at the Edge | Aerospace Track